5.6 Numerical Methods and Invariant Distributions

In the previous section, we introduced the temporal average of a function along a solution of a differential equation (as a means of computing the microcanonical ensemble average). Since we cannot compute an exact trajectory, we would hope that such an average could be computed in an analogous way using a numerical method. If z_n represents the approximation to $z(t_n)$, then we might attempt to estimate $\bar{g}_\Gamma$ by

$$\bar{g}_\Gamma \approx \lim_{n\to\infty} n^{-1} \sum_{k=0}^{n} g(z_k).$$

However, as is well known, numerical methods may introduce growing errors so an infinite average might not make sense, even if the time average does! Still we might hope that trajectory averages could be approximated by discrete averages of moderately long length. In the several examples in this chapter, we have used numerical methods to simulate Hamiltonian dynamics in order to gain intuition about their ergodic properties, but we have not discussed the effect of the choice of the discretization method itself on this issue.

We have seen that the Liouville equation defines the propagation of the distribution associated to system of differential equations. In a similar way, a numerical method $z_{n+1} = \mathscr{F}_h(z_n)$ induces an effective distributional propagator. One way to understand this propagator is to interpret the numerical method as being approximately equivalent to the solution of a perturbed differential equation

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{f}(z) + h^r \boldsymbol{f}_{(r)}(z) \equiv \tilde{\boldsymbol{f}}_h(z).$$

The perturbation term may be viewed as the finite truncation of the perturbative expansion obtained from the backward error analysis using the methods of the previous chapter. (r is the order of the numerical method.) Let us assume that the perturbed equation is a realistic model for the numerical solutions, then define the evolving density by

$$\frac{\partial \rho}{\partial t} = \mathcal{L}^\dagger_{\tilde{f}} \rho \stackrel{\text{def}}{=} \mathcal{L}^\dagger_f \rho + h^r \mathcal{L}^\dagger_{f_{(r)}} \rho.$$

Under suitable boundedness assumptions on the remainder, we could expect that the distributional error remains bounded by a quantity of order h^r. Unfortunately, it is not always true that ρ remains bounded.

As a simple illustration, consider the harmonic oscillator with Hamiltonian $H(q,p) = p^2/2 + q^2/2$ and the invariant distributions obtained using several numerical methods. Applying Forward Euler to the harmonic oscillator, we have

$$q_{n+1} = q_n + hp_n, \qquad p_{n+1} = p_n - hq_n,$$

or

$$\begin{bmatrix} q_{n+1} \\ p_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & h \\ -h & 1 \end{bmatrix} \begin{bmatrix} q_n \\ p_n \end{bmatrix}$$

and we see that the eigenvalues of the propagation matrix are outside the unit disk. Thus both solution components tend to infinity unless $(q_0, p_0) = (0, 0)$. This means that with the single exception $\rho(q, p, 0) = \delta[q]\delta[p]$ there is no limiting (invariant) distribution for Forward Euler. To put it another way, the only solution, in the weak sense, of $\mathcal{L}_{\tilde{f}}^{\dagger}\rho = 0$ is the point distribution $\delta[q]\delta[p]$, and this solution is not "asymptotically stable."

As an alternative, consider the Backward Euler method: $q_{n+1} = q_n + hp_{n+1}$, $p_{n+1} = p_n - hq_{n+1}$. Then it is easily shown that both eigenvalues of the matrix lie in the interior of the unit disk, and hence all solutions of the recurrence relation tend to the origin with increasing n (the origin is an attractive equilibrium point). In this case, all densities evolve toward the Dirac distribution centered at the origin: $\delta[q]\delta[p]$. The only (distributional) solution of $\mathcal{L}_{\tilde{f}}\rho = 0$ is again $\delta[q]\delta[p]$, which in this case is attractive.

Now consider the application of the Symplectic Euler method, $q_{n+1} = q_n + hp_{n+1}$, $p_{n+1} = p_n - hq_n$. As we know, the solution through any given initial point remains confined to an ellipse for all n defined by $\tilde{H}_h(q, p) = \tilde{H}_h(q_0, p_0)$, where $\tilde{H}_h$ is the perturbed Hamiltonian defined by the backward error analysis. This implies that a density which is a function of $\tilde{H}_h$ will remain invariant under the associated Liouville equation. There are many solutions of $\mathcal{L}_{\tilde{f}}^{\dagger}\rho = 0$, none of which is attractive.

The harmonic oscillator is of course an integrable system. When the underlying system is a chaotic Hamiltonian system, the trajectories from any collection of initial conditions become disordered (even while preserving the energetic constraints defined by their initial states). The hope (in some cases, naive) is that typical trajectories computed using a symplectic method applied to such a chaotic system provide space filling curves on the $\tilde{H}_h = \mathrm{E}$ perturbed energy surface.

Example 5.7 (Ergodicity in the Trimer) At positive energies, the trimer is unstable and the atoms separate, into a bound pair and a single atom. At very low (but admissible) energies, the timer has an insurmountable energy barrier between the two basins. At moderate energies, the trimer will visit both basins, however, the mean passage time between basins may be greatly extended depending on the energy level.

Consider the level set of potential corresponding to $U = -0.5$. Note that the kinetic energy is everywhere positive, so that a trajectory with total energy $\mathrm{E} = -0.5$ will have to be bounded within the $U = -0.5$ level set. Would the dynamics (started from an initial condition with total energy $\mathrm{E} = -0.5$) completely fill in this level set over time? Because this system is chaotic, we cannot write down a formula for the exact solution, or find exact trajectories of the system, but we can use numerical methods to attempt to shed some light on the question. In Fig. 5.8 we